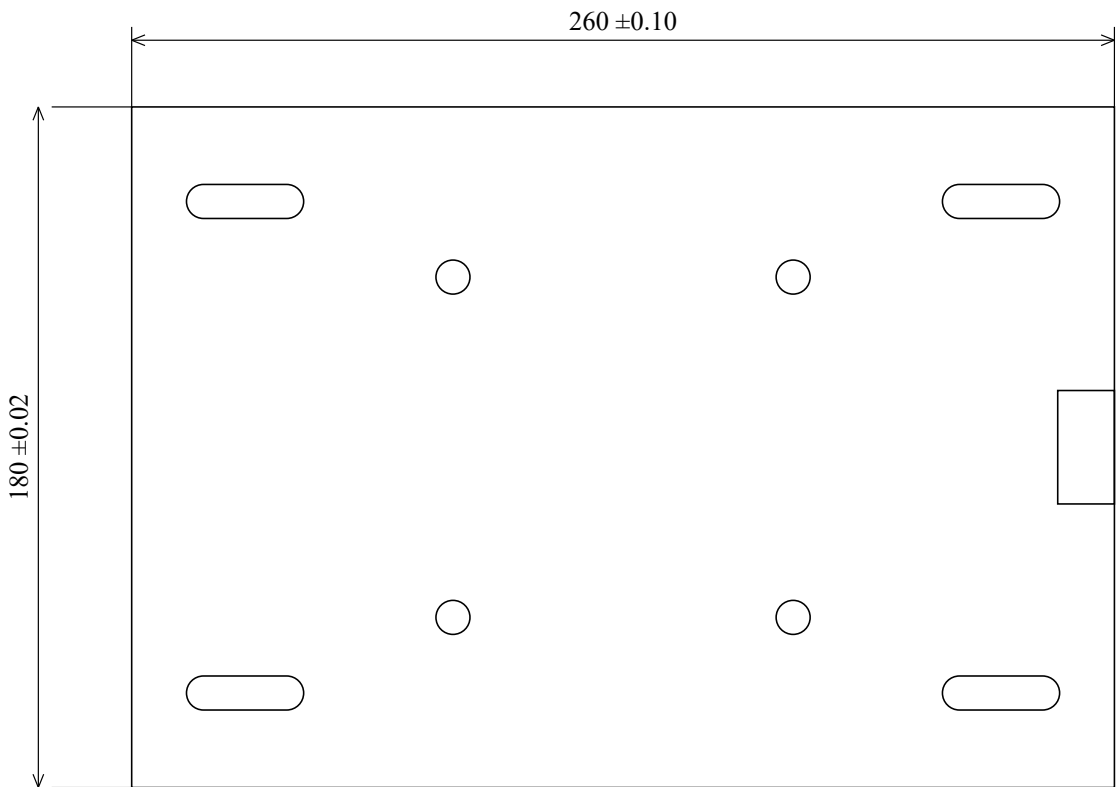
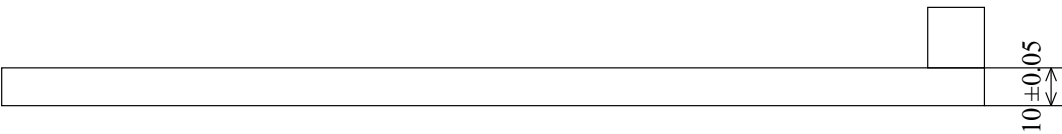


Motor plate and functional datums

REPORT DESIGN R1 | 8 September 2026 | Dimensions in mm | Third angle views | Do not scale



TOP VIEW 1:2 | origin at left lower corner on datum A



FRONT VIEW 1:2 | A = bottom plane | B = long side | C = left end

ANALYSIS LINK

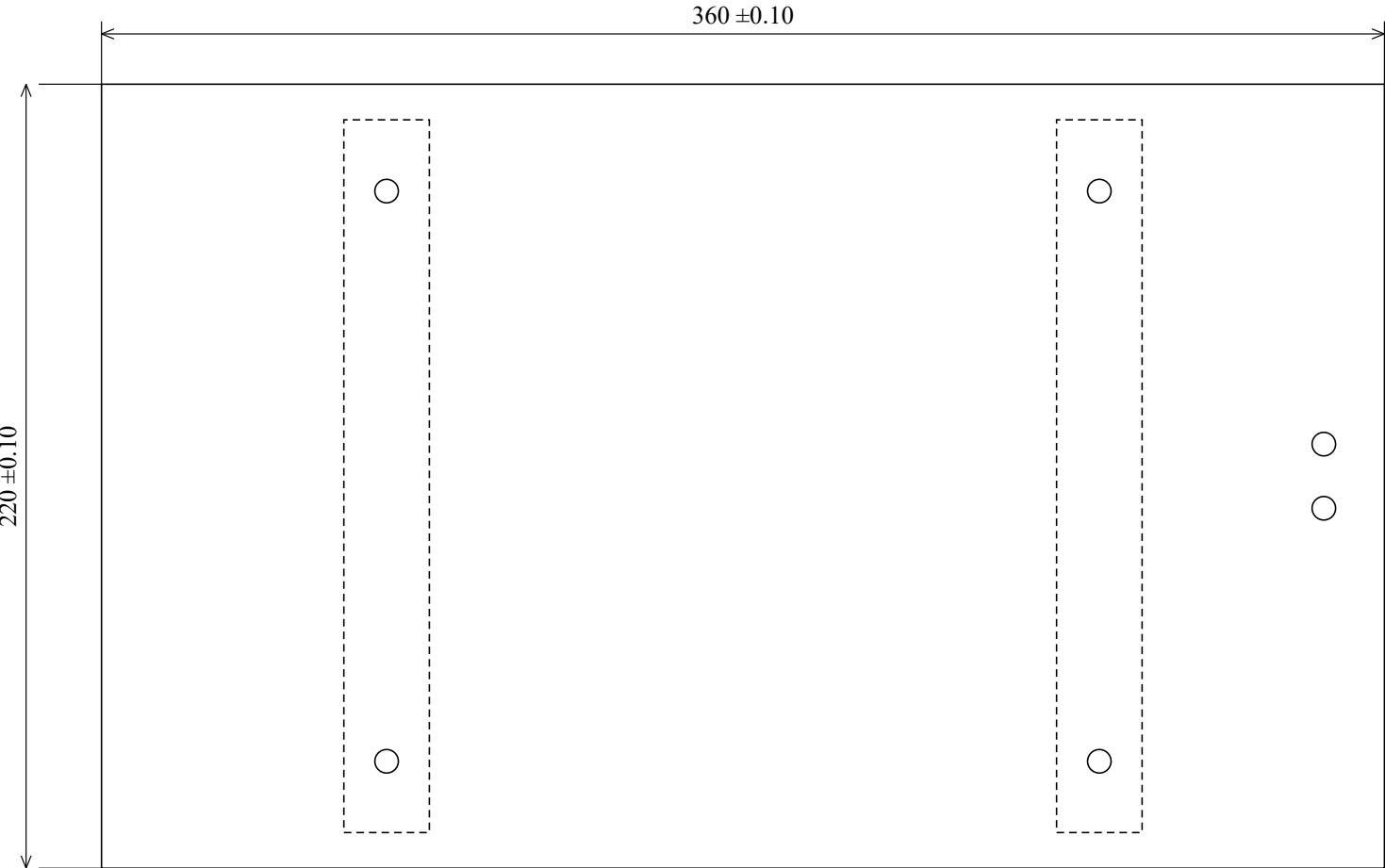
Plate body mesh includes four slots and four motor holes.
Peripheral boss omitted from locked-service mesh; see A4.
Support centres x = -100 and +100 give the 200 mm span.
New CAD dimensions are R1 report decisions.

PART PR-PLT-001 | QTY 1 | 6061-T6
Machine body and integral thrust boss from stock at least 27 thick.
Boss: x = 245 to 260; y = 75 to 105; z = 10 to 26.
4 slots: 31 ±0.10 overall × 9 +0.10/-0.05; ends R4.5.
Slot centre coordinates from C and B:
(30,25), (30,155), (230,25), (230,155).
Slot length along x. X coordinates ±0.10; Y coordinates basic.
4 motor holes: diameter 9 +0.10/0 THRU.
Motor hole basic coordinates: (85,45), (85,135),
(175,45), (175,135). Motor pattern 90 × 90.
Motor hole position: cylindrical zone diameter 0.10 | A | B | C.
Slot median-plane position: zone width 0.20 | A | B | C.
A flatness 0.05; B perpendicular to A within 0.05.
C perpendicular to A within 0.05.
Deburr edges 0.2 to 0.5; no burrs on seating faces.
General linear tolerance ±0.10 unless noted.
Feature controls below use vector-drawn symbols.
Basic motor coordinates are defined by the position control.

Motor hole position		Ø0.10	A	B	C
Slot median plane		0.20	A	B	C
Datum A flatness		0.05			

Base support rails and side guides

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BASE TOP 1:2 | 360 × 220 × 12 steel | global centre at (0,0)



RAIL HOLE MAP 1:2 | view rotated for space



GUIDE TOP 1:2 | all parts deburred 0.2 to 0.5 | general ±0.10

PR-BAS-001 | QTY 1

6 diameter 6.6 THRU for M6 screws from underside.

Global coordinates: (±100, ±80) and (163, ±9).

Bottom counterbores diameter 13 × 8 deep for M6 heads.

Base attaches to the machine through a clamped interface.

External frame anchors are outside this mount design.

Thickness 12 ±0.10; seating flatness 0.10.

PR-RAL-001 | QTY 2

24 × 200 × 20 steel; rail global centres x = ±100.

2 M8 × 1.25 THRU at local x = 12, y = 35 and 165.

2 M6 × 1 blind 15 from bottom at x = 12, y = 20,180.

2 M5 × 0.8 blind 12 from top at x = 12, y = 4.975,195.025.

Both top seating surfaces coplanar within 0.05.

PR-GDE-001 | QTY 2

260 × 9.95 × 16 steel; 2 diameter 5.5 THRU.

Hole centres x = 30,230; y = 4.975 from strip edges.

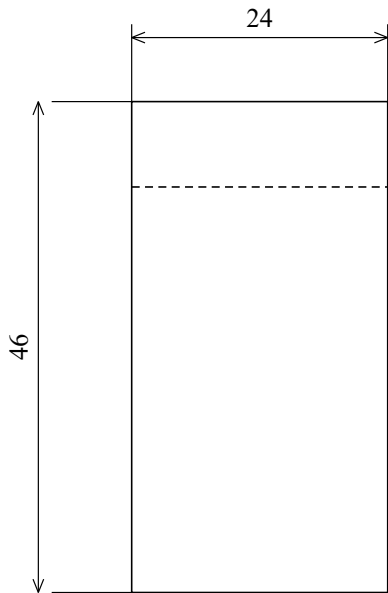
Install clear distance between guides 180.10 +0.05/0.

Guide inner faces parallel within 0.03 over 260.

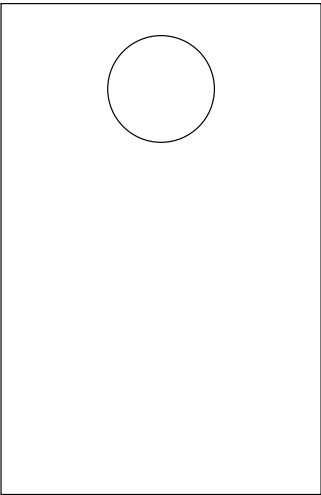
Guide hole clearance allows setting before tightening.

Adjuster and locking hardware

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REACTION BLOCK SIDE | enlarged view



END VIEW | tapped screw axis 38 above base

LOCKED SERVICE LOAD PATH

4 M8 × 30 class 8.8 clamp screws; 24 OD × 9 ID × 3 hardened washers.

Nominal screw engagement in rail = 17 mm, with washer upper face at z = 13.

4 M8 × 35 motor screws; 4 nuts; 8 plain washers (16 OD × 8.4 ID × 1.6).

6 M6 × 16 base screws; counterbored heads. 4 M5 × 25 guide screws.

Threads, fastener heads, nuts and small chamfers are simplified or omitted in the STEP envelopes.

Estimated clamp preload is a calculation input. Select installation torque using verified joint friction.

Support-rail and base stiffness, thread pullout and fatigue require separate release checks.

PR-ADJ-001 | QTY 1 | steel

Block size 24 × 30 × 46; global x = 151 to 175.

Bottom z = -20; top z = 26; y = -15 to +15.

M10 × 1.5 through along x; axis y = 0, z = 18.

2 M6 × 1 blind 15 from bottom.

Hole locations in bottom view: x = 12; y = 6,24.

General ±0.10; perpendicularity of screw axis to boss thrust face within 0.10 over 30.

PR-HW-004 | M10 × 1.5 × 60, class 8.8

Flat end bears on integral plate boss; one locknut.

Nominal screw spans x = 130 to 190.

At travel extremes the screw tip is 120 or 140.

Thread engagement through 24 mm block is retained.

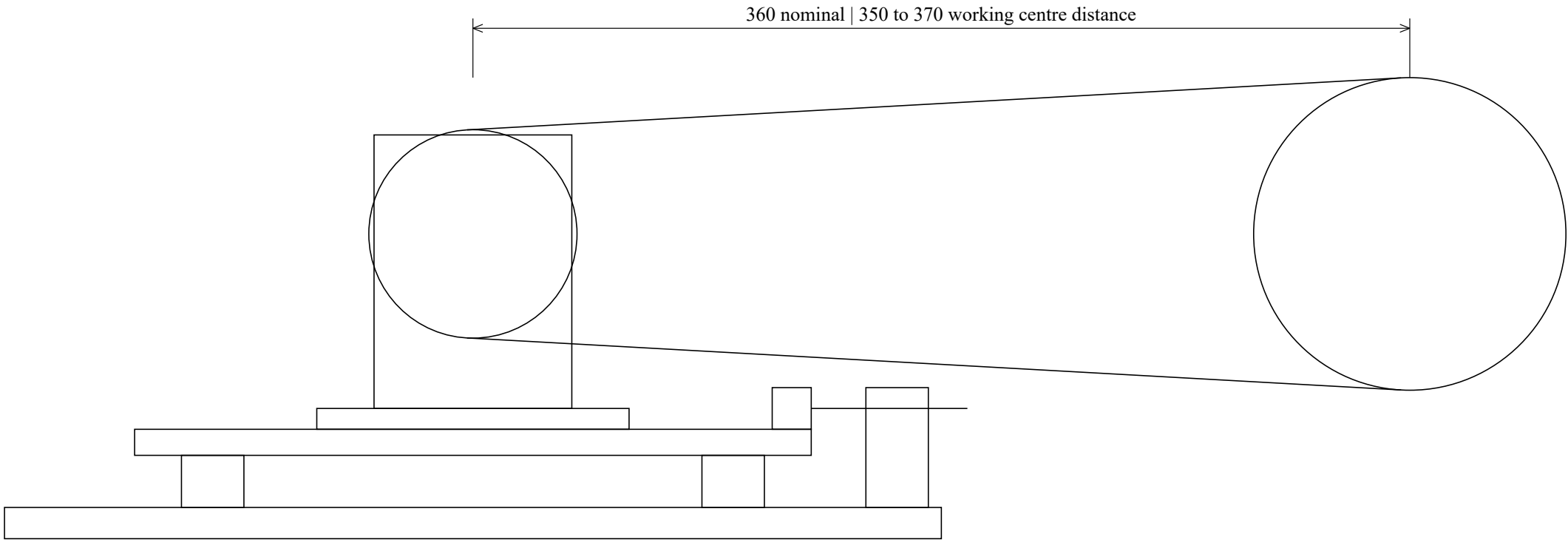
Screw free length from block to boss: 11 to 31.

One turn gives 1.5 mm travel; 20 mm = 13.33 turns.

The screw pushes left to increase tension; belt pull or manual movement returns the plate while clamps are loose.

Assembly travel belt envelope and alignment setting

REPORT DESIGN R1 | 8 September 2026 | Dimensions in mm | Third angle views | Do not scale



SIDE ELEVATION | pulley axes along y | pitch plane y = 85 | drive shaft z = 85

ASSEMBLY AND SETTING

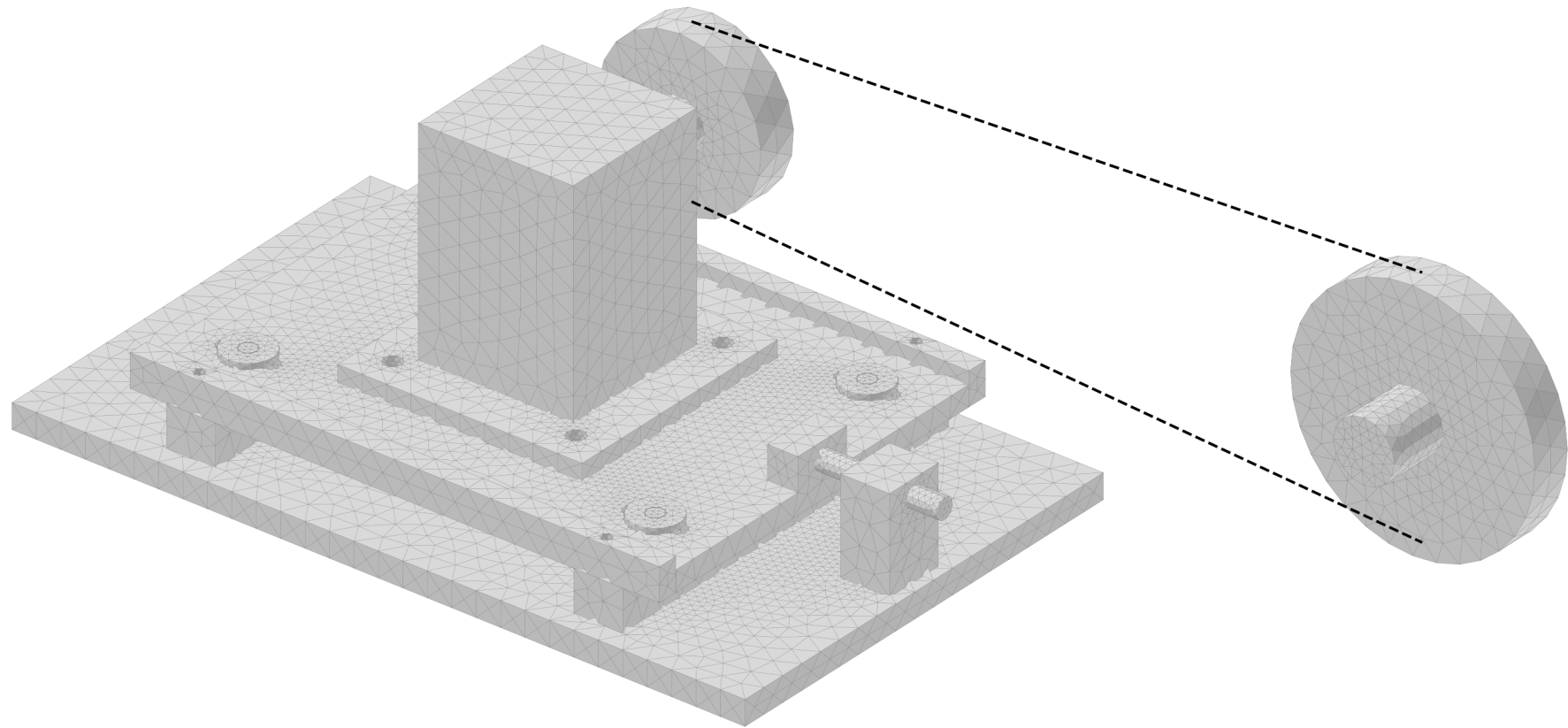
- Loosen four plate clamps. Turn M10 screw to move plate x = -10 to +10 from nominal.
- 31 mm slots around 8 mm shanks give 23 mm geometric travel; reserve 1.5 mm per end.
- Nominal washer edge clearance at worst travel is 8 mm; guide side clearance is 0.08 to 0.17 total.
- Motor body envelope 76 × 76 × 105 above 120 × 120 × 8 foot; four holes on 90 × 90 pattern.
- Drive pulley pitch radius 40; driven radius 60; both faces 20 wide. This is a pitch-envelope design.
- Pitch lengths at centre distances 350/360/370 are 1015.30/1035.27/1055.24 mm.
- Belt profile, standard length, rated capacity and guard are selected with the actual drive supplier.
- Driven shaft support and machine frame are external interfaces, not parts of this mount BOM.

AXIAL ALIGNMENT

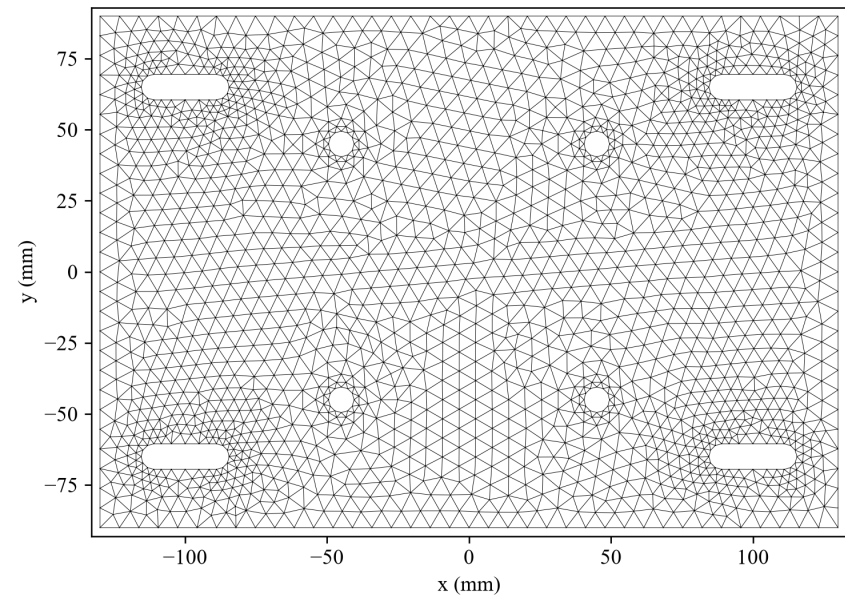
- Drive pitch plane nominal y = 85.
- Provide pulley hub setting y = 84 to 86.
- The ±1 range exceeds the ±0.28 stack.
- Measure at opposing pulley rim points.
- Accept static readings within ±0.15 mm.
- Budget uncertainty 0.03 and lock shift 0.02.
- Recheck after all fasteners are tightened.
- Record the actual reading and instrument.
- This sheet defines the inspection requirement;
- it does not record an inspection pass.

CAD assembly and finite element mesh views

REPORT DESIGN R1 | 8 September 2026 | Dimensions in mm | Third angle views | Do not scale



ISOMETRIC VIEW | generated from PR-ASM-001_R1.step | pitch belt shown as dashed tangent lines



GEOMETRY AND MODEL TRACEABILITY

Assembly shows plate, integral boss, base, rails, guides, reaction block, clamp washers and motor/drive interfaces. Threads and small hardware details are simplified. Driven support is an external interface, not a floating design part.

Mesh view shows actual top boundary triangles of the plate body. Final solid mesh: 31,378 tetrahedra; 168,708 displacement DOFs. Quadratic displacement functions are used on straight-sided tets. The peripheral boss and hardware are excluded from this mesh.